Clinic No-Show Problem Analysis

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1. Situation

Public clinics in Vitória, Brazil lose around 3,500 booked appointments a week to no-shows. Patients booked a few days ahead get an SMS reminder, but the team believes it doesn't help, because patients who got an SMS miss more often (27.6% vs 16.7%). Before dropping, keeping or extending reminders, the clinic needs to know whether they work and whether they are actually being sent. The analysis uses the clinics' real appointment records (110,527 appointments, April to June 2016), with public holiday and weather data.

2. Current workflow and evidence

Step	Current workflow	Evidence or issue
1	Patient books a slot at the desk or by phone	35% are same-day bookings; only 4.7% of these are missed
2	Slot is saved in the scheduling system	Booking time is labelled UTC but matches local clinic hours
3	**SMS reminder is sent**	**Only if booked 3+ days ahead; no SMS at all on 7 days (May 9, 13, 16-20)**
4	Appointment day	No time of day recorded; May 23 and 27 have no appointments
5	**Patient attends or does not show**	**28.5% of advance bookings are missed**
6	Patient rebooks	5% of patients account for 35% of all no-shows

Main bottleneck. The no-show at step 5, driven by how far ahead the slot is booked (step 1) and whether the reminder goes out (step 3). No-shows rise from 4.7% for same-day bookings to 33% for bookings 31+ days ahead.

3. Stakeholders

Stakeholder	Goal	Constraint	Key question
Operations manager (owner)	Fewer wasted slots	Fixed budget, no new staff	Is the SMS worth paying for?
Reception staff	A short call list, not a report	Little spare time per desk	How many calls a day can each desk make?
Doctors and nurses	A full clinic day	No double-booking without a rule	Would overbooking be acceptable?
Patients	Care on time, easy rebooking	Phone access unknown	Is the reminder clear and does it arrive?
SMS vendor / city IT	A clear sending rule	No one watches daily send volume	Why did 7 days send nothing?

4. Evidence map

Item	Current view	Status	How to verify
SMS sending rule	Never sent when booked under 3 days ahead	MEASURED	0 of 50,501 rows; confirm with vendor
Reminder effect	**SMS patients miss 5.4 pts less at the same lead time**	**MEASURED**	**Outage days give 4.3 pts; confirm in a controlled test**
Weather	Too few rainy days (2) to tell	WEAK	18.9% rainy vs 20.3% dry; recheck later
Same-day bookings	Mostly walk-ins	ASSUMPTION	Ask reception how same-day slots are made
Reminder outage	Vendor failure or export gap	UNKNOWN	Vendor send logs for May 9-20
Patients with no SMS	18% of eligible patients on normal days; no phone?	UNKNOWN	Check the patient register
Empty weekdays	May 23 and 27 closed, or missing from the export	UNKNOWN	Ask managers; only May 26 is a holiday

5. Problem statement

Advance-booked appointments at Vitória's public clinics end in a no-show 28.5% of the time. The clinic reads its data as "SMS reminders don't work", but reminders only go to patients booked 3+ days ahead, who miss most often anyway, and on 7 days no reminders were sent at all. The clinic needs a reliable, repeatable view of whether reminders reduce no-shows and whether they are delivered, so it can decide to drop, fix or extend them.

6. KPI and supporting metrics

Role	Metric	Baseline (6 weeks)	Decision trigger
KPI	**No-show rate of appointments booked 1+ days ahead**	**28.5%**	**A week above 30%**
S1	Reminder coverage of eligible appointments (booked 3+ days ahead)	59.1% (81.7% normal days)	A day below 50%, or a week below 70%
S2	Reminder effect at the same lead time	5.4 points	Below 1 point: don't scale SMS
S3	Share of no-shows from patients with 2+ no-shows	35.3%	Sizes the reception call list
S4	Advance bookings made 15+ days ahead	38.6%	Context for slot policy

7. MVP scope and success

Area	Scope
In scope	A weekly pipeline (retrieve, validate, model, metrics), a daily alert when reminder coverage falls below 50%, and a weekly call list of patients with 2+ earlier no-shows
Out of scope	No-show prediction or ML, overbooking rules, SMS wording, changing vendor, cost figures
Success	In a 4-week test against a control group (fix outages, send SMS to 1-2 day bookings, call repeat no-shows), the KPI falls by at least 2 points (to 26.5% or lower), coverage stays above 80% every day, and opt-outs do not rise

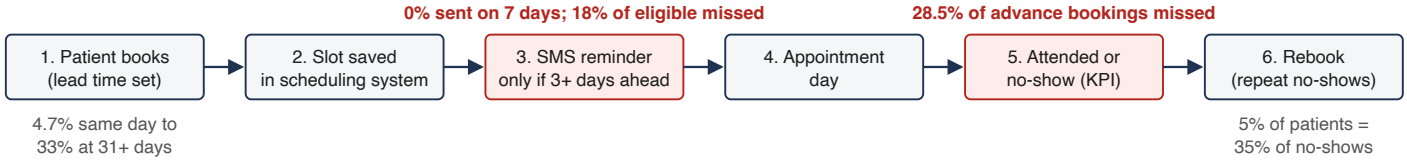
8. Why this is the right problem to solve first

I listed every problem the data showed and compared them before building anything. I did not start with a prediction model or an AI tool: the first question was which problem is worth fixing, and whether the data can support it.

Candidate problem	Size (6 weeks)	Can we act on it now?	Decision
SMS reminders judged "not working", and not sent on 7 days	**60,021 eligible bookings; 24,539 got no SMS**	**Yes: the SMS already exists, the fix is a daily check**	**Solve first**
Repeat no-show patients	5% of patients, 35% of no-shows	Yes: a short call list for reception	Part of the MVP
Long lead times	38.6% of advance bookings are 15+ days out	Slowly: needs a slot policy change	Later
Rain / weather	18.9% vs 20.3%, only 2 rainy days	No: can't control it, weak evidence	Not now
No-show prediction model	no send times, no clinic or doctor fields	No: basics are unreliable first	Out of scope

Why reminders come first. (1) The clinic is about to decide from a misleading number: the naive comparison says SMS makes things worse, so they may stop paying for it. (2) It is cheap: the reminder is already paid for, and a daily coverage alert costs almost nothing. (3) It is big: reminding the 24,539 missed patients would have meant about 1,325 fewer no-shows, about 1.8 KPI points. (4) The other fixes depend on it: a call list or an overbooking rule can't be judged if reminders fail silently.

9. Where the problem sits in the workflow



10. Evidence for the main conclusion

The naive view: 27.6% no-show with an SMS vs 16.7% without. But SMS only goes to bookings 3+ days ahead, and those miss more anyway. Compared at the same lead time, the picture flips. The outage days give a second check, because on those days nobody got an SMS whoever they were.

Lead time	Appointments	With SMS	Without SMS	Gap	Outage-day gap
3-7 days	20,247	23.8%	26.6%	2.8 pts	1.9 pts
8-14 days	12,025	28.1%	33.8%	5.7 pts	5.8 pts
15-30 days	17,371	29.8%	36.9%	7.1 pts	5.5 pts
31+ days	10,378	30.2%	37.5%	7.3 pts	5.2 pts
Weighted	**60,021**			**5.4 pts**	**4.3 pts**

Both checks point the same way: reminders are linked to about 4-5 points fewer no-shows. It is not a randomised test, so I don't claim it as proven; the next step is a controlled 4-week trial.

11. Facts, assumptions and unknowns

Facts (measured in the data)	Assumptions (mine, to confirm)	Unknowns (need the client)
110,527 appointments; only 6 rows fail checks	Booking times are local, although marked UTC (peak at 07h)	Why no SMS went out on May 9, 13, 16-20
No SMS ever sent under 3 days ahead (0 of 50,501)	Same-day bookings are mostly walk-ins	Whether "SMS received" means sent or delivered
7 appointment days had 0% SMS coverage	Neighbourhood is where the clinic is	Why 18% of eligible patients get no SMS (no phone?)
May 23 and 27 are empty; only May 26 is a holiday	Trigger levels are my own, from 6 weeks	Whether a cancellation is recorded as a no-show

12. What I built and what I recommend

Item	Detail
Data	Real public data, nothing synthetic: Kaggle "Medical Appointment No Shows" CSV (checksum-pinned), Nager.Date holidays API, Open-Meteo weather API
Pipeline	retrieve → validate (12 rules, rows flagged not deleted) → SQLite model → metrics in SQL; logging, retries, cached fallbacks, and the run is held (exit 2) if over 2% of rows fail
Recommend	Keep the SMS; alert when a day's coverage is under 50%; ask the vendor about May 9-20; test SMS for 1-2 day bookings plus reception calls for repeat no-shows
Limits	Only 6 weeks of 2016 data; outage check is not randomised; no clinic, doctor or time-of-day fields; only national holidays checked
Links	Repo: github.com/tek-wizard/FDE-assignment · Demo video: loom.com/share/54ed45edf2b348ea987dac63c069a418